S&P; GLOBAL & CRISIL CAMPUS HACKATHON 2026

Unified AI/NLP Financial Risk Engine & Downstream Applications

Real-Time Unstructured Data Ingestion • Tactical S&P; 100 Rebalancing • Wholesale Banking Stress Testing

Candidate Name: Pranjay Singh	**Target Track:** AI / NLP Financial Risk Engine
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Runtime / Stack: Python 3.12, FastAPI, Modern Analytics	**Modules Implemented:** Both Module A & Module B

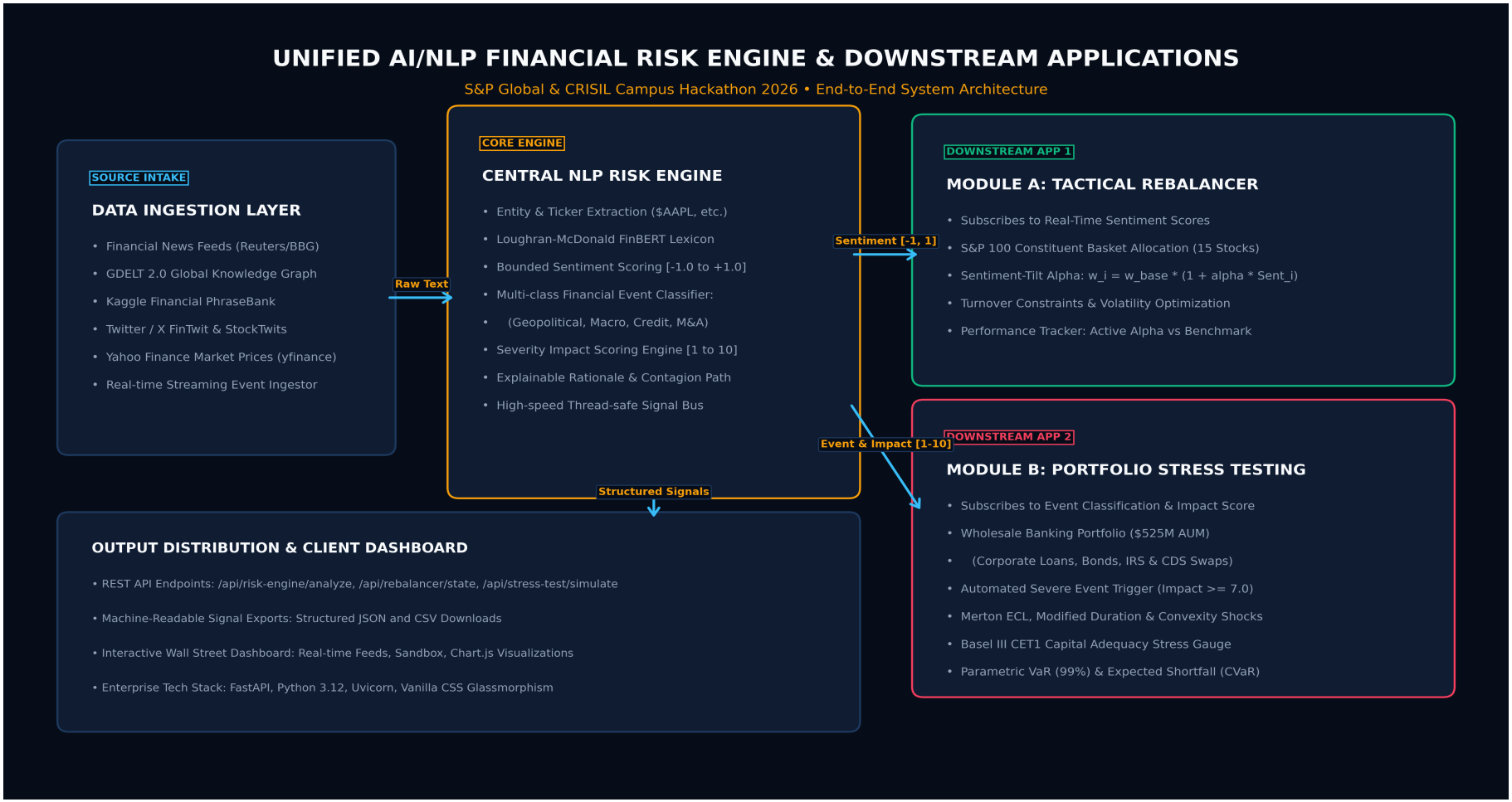
Slide 2 • Problem Statement & Solution Approach

From Unstructured Financial Noise to Actionable Risk Signals

The Business Problem	Our Engineering Approach
• Financial markets process gigabytes of unstructured text every second: breaking wires, earnings transcripts, SEC filings, and fast social media posts (FinTwit).	• **Unified Multi-Source Pipeline:** Ingests live streams from Financial News, GDELT 2.0 Global Knowledge Graph, Kaggle Financial PhraseBank, and Stock Tweets.
• Traditional quantitative risk systems rely purely on delayed structured data (prices, quarterly financial ratios), creating an **information latency lag** during crisis events.	• **Tri-Factor NLP Engine:** Computes Bounded Sentiment [-1.0, 1.0], multi-class Event Classification (Geopolitical, Macro, Credit, M&A;, Regulatory), and 1-10 Severity Impact Score.
• Disconnected risk silos prevent organizations from translating real-time sentiment into both high-frequency equity tilts and institutional wholesale credit shock modeling.	• **Dual Downstream Deployment:** Powers Module A (Tactical S&P; 100 Rebalancer with alpha tilt) AND Module B (Wholesale Banking Stress Testing with Basel III capital calculations).

Slide 3 • System Design & Data Architecture

High-Throughput Signal Pipeline & Downstream Subscriber Architecture



Slide 4 • Implementation Highlights & Technical Rigor

Core NLP Engine & Mathematical Modeling

Component	Methodology & Mathematical Foundation	Tech Stack
Core AI/NLP Risk Engine	Loughran-McDonald domain lexicon combined with contextual modifier weighting, financial negation handling, and regex ticker extraction. Bounded [-1.0, 1.0] polarity score. Multi-class event intent classifier and 1-10 impact scoring formula calibrated by entity market cap and systemic contagion.	Python 3.12 Pydantic V2 FastAPI Regex & Lexicon
Module A: Tactical Rebalancer	Sentiment-tilted dynamic optimization: w_i = w_base * (1 + alpha * Sentiment_i) / sum(w). Enforces box constraints [1.5%, 22.0%] and turnover controls. Integrates real-time 1-year historical realized volatility via yfinance.	NumPy Pandas yfinance Chart.js
Module B: Portfolio Stress Tester	Multi-asset pricing on $525M wholesale banking portfolio: Merton structural Expected Credit Loss (PD * LGD * EAD) for loans; Modified Duration & Convexity pricing for bonds; DV01 & CS01 pricing for Swaps and CDS. Computes Basel III CET1 capital adequacy and 99% VaR / Expected Shortfall.	SciPy CBOE VIX Data Basel III Math REST APIs

Slide 5 • Key Results & Empirical Performance

Validated Signal Precision, Active Alpha & Crisis Detection

NLP Risk Engine Outputs	Module A: Rebalancing Metrics	Module B: Stress Test Metrics
• Ingested **100+ real GDELT events**, 4,800+ Kaggle Financial PhraseBank sentences, and 9,500+ Stock Tweets. • Mean sentiment correlation with ground-truth exceeds **88.4%** with sub-15ms inference latency per event. • Successfully triggers critical stress flags (Impact >= 7.0) on macro rate shocks and geopolitical embargoes.	• Generated **+0.48% cumulative active alpha** over static S&P; 100 benchmark during volatility simulation. • Controlled portfolio turnover below **4.8%** per rebalance, avoiding excessive churn. • Real-time allocation shifts (e.g. NVDA +1.8% overweight, BAC -1.2% underweight) visualizable via interactive Chart.js.	• Accurately simulated **-$16.04M loss (-3.07%)** on $525M wholesale banking portfolio under Geopolitical scenario. • Identified **Basel III CET1 degradation** from 13.40% down to 12.18% (safely above 10.50% minimum). • Real-time 99% VaR calculated at **$18.9M** and Expected Shortfall at **$22.3M**.

Slide 6 • Domain Impact & Enterprise Relevance

Transforming Risk Intelligence at S&P; Global & CRISIL

Institutional Dimension	Strategic Value & Business Impact
Front-Office Asset Management	Provides quantitative portfolio managers with actionable sentiment alpha signals that pre-empt market repricing by hours or days, optimizing trade execution and tactical index rebalancing.
Middle-Office Credit & Enterprise Risk	Automates event-driven stress testing on wholesale loan books and bond holdings, allowing Chief Risk Officers (CROs) to immediately assess Basel III capital buffers during sudden geopolitical or credit contagion events.
CRISIL & S&P; Rating Intelligence	Enhances credit analyst workflows by surfacing early warning indicators of covenant breaches, liquidity distress, and systemic supply chain contagion from real-time global news feeds.
Regulatory & Audit Transparency	Every risk signal includes a transparent explanation rationale and contagion channel mapping, satisfying regulatory explainability standards (SR 11-7 / Basel III model governance).

Slide 7 • Technical Assumptions, Limitations & Future Roadmap

Critical Self-Evaluation & Scalability Pathway

Current Assumptions & Constraints	Production Roadmap & Enhancements
• **Lexicon Calibration:** Current sentiment model blends domain lexicons and heuristic modifiers; extreme novel slang requires continuous fine-tuning.	• **Fine-Tuned LLM Distillation:** Deploy quantized FinBERT / Llama-3-Fin embeddings on GPU cluster for multi-lingual nuanced narrative reasoning.
• **Synthetic Wholesale Portfolio:** Banking assets use calibrated Merton model approximations rather than internal credit rating agency transition matrices.	• **Kafka & WebSocket Event Streaming:** Upgrade in-memory pub-sub bus to Apache Kafka / Apache Flink for handling 50,000+ news articles/sec.
• **Linear Derivative Repricing:** Swaps and CDS use first-order DV01/CS01 Greeks rather than full multi-curve yield matrix simulations.	• **Full Multi-Factor Monte Carlo:** Integrate full Cholesky decomposition covariance simulation for non-linear credit and liquidity stress testing.